

TripMate AI: Collaborative Group Activity Planning

Powered by an Autonomous Multi-Agent AI Architecture

Project Idea & Technical Concept Proposal for Supervisor

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1. Project Overview & Core Idea

1.1. The Elevator Pitch

TripMate AI is an intelligent, real-time collaborative web application engineered to eliminate the friction, indecision, and organizational fatigue inherent in group decision-making. Whether coordinating multi-day vacations, finding the perfect dinner spot, organizing a cafe hangout, or planning entertainment outings, TripMate AI transforms chaotic discussions into an actionable, consensus-driven plan in **under 15 minutes**.

1.2. The Core Paradigm: Democratic Human-in-the-Loop AI

Unlike standard AI chatbots that generate disconnected text responses, TripMate AI integrates an autonomous **Multi-Agent AI System (LangGraph Python)** directly into a structured collaborative planning pipeline:

1. Event Creation → 2. AI/Manual Proposal → 3. Democratic Voting → 4. Owner Confirmation → 5. Debt Settlement

The platform operates on a strict **Human-in-the-Loop** principle: the AI acts as an expert research assistant producing structured, modifiable DRAFT plans, while humans retain complete democratic authority through voting, granular modifications, and owner sign-off.

1.3. Key Value Propositions

- **Autonomous Multi-Agent Intelligence:** Specialized sub-agents automatically search live locations via Google Places, extract verified restaurant menus and activity pricing, compute optimal route sequences, and balance budgets.
- **DeepSeek-V4 Next-Gen Reasoning:** Integrates the latest **DeepSeek-V4** model family, utilizing high-throughput tool calling for data retrieval and deep reasoning tokens to resolve split-vote deadlocks among members.
- **Automated 3-Table Financial Settlement:** Eliminates post-trip financial headaches with an algorithm that tracks pool advances vs. actual spending and computes the minimum peer-to-peer bank transfers needed.

2. Problem Statement & Motivation

Group planning among 3 to 10 participants consistently breaks down across four major pain points:

1. **Endless Debates & Information Fragmentation:** Casual questions like *"Where are we eating tonight?"* spawn hundreds of unstructured messages across messaging apps (Zalo, Messenger), causing crucial details to get buried.
2. **Single Organizer Overload:** One person typically bears the entire burden of reading reviews, checking opening hours, comparing hotels, estimating transit times, and individually polling each friend.

- 3. **Voting Deadlocks & Conflicting Preferences:** When members have opposing tastes (e.g., budget vs. luxury, Japanese cuisine vs. BBQ), tied votes cause inertia and frequently result in canceled plans.
- 4. **Post-Event Expense Splitting Friction:** Tracking who paid for specific stops and manually calculating uneven debts creates social awkwardness and accounting errors.

Table 1: Comparative analysis: TripMate AI vs. existing market solutions

Planning Tool	Core Limitations	TripMate AI Solution
Group Chats (Zalo/Messenger)	Unstructured, noisy, no poll integration	Structured timeline with interactive voting
Shared Docs (Notion/Sheets)	Static, manual data entry, no live data	Autonomous live place lookups & menu retrieval
Generic AI (ChatGPT/Claude)	Prone to hallucination, single-user only	Real-world Google Places API + multi-user context
Traditional Travel Apps	Limited to multi-day trips, no conflict AI	Supports 6 event types + AI conflict resolution

3. System Architecture & Collaborative Workflow

3.1. End-to-End Collaborative Lifecycle

TripMate AI formalizes group planning into five seamless stages:



Figure 1: End-to-end collaborative planning workflow

3.2. Multi-EventType Support: Beyond Just Vacations

Rather than focusing exclusively on multi-day travel, TripMate AI optimizes search heuristics and UI cards for six distinct daily group scenarios:

- **1. Dining (DINING):** Curates restaurants, extracts signature dishes/menus, checks dietary restrictions, and provides price per person.
- **2. Cafe & Catch-ups (HANGOUT):** Finds coffee shops filtered by vibe (quiet for work, aesthetic photo spots, rooftop lounges).
- **3. Entertainment (ENTERTAINMENT):** Suggests recreational spots (karaoke, bowling, escape rooms, arcade) with ticket costs and capacity limits.
- **4. Sightseeing (SIGHTSEEING):** Identifies cultural landmarks, museums, and exhibitions with operating hours and parking tips.
- **5. Multi-Day Travel (TRAVEL):** Sequences multi-day itineraries, optimizes travel routes, and compares accommodations.
- **6. Custom Events (CUSTOM):** Provides open-ended collaborative planning for birthdays, picnics, or team-building events.

4. Next-Gen AI Multi-Agent Architecture (DeepSeek-V4)

4.1. Orchestrator–Worker Pattern on LangGraph Python

The AI architecture utilizes an **Orchestrator–Worker Pattern** deployed directly within the Python backend:

4.2. DeepSeek-V4 Dual-Engine Strategy

- **DeepSeek-V4 Conversational Engine:** Executes intent classification, multi-lingual natural language chat, and high-frequency tool calling (Google Places, OpenWeatherMap) with exceptional throughput and minimal latency.

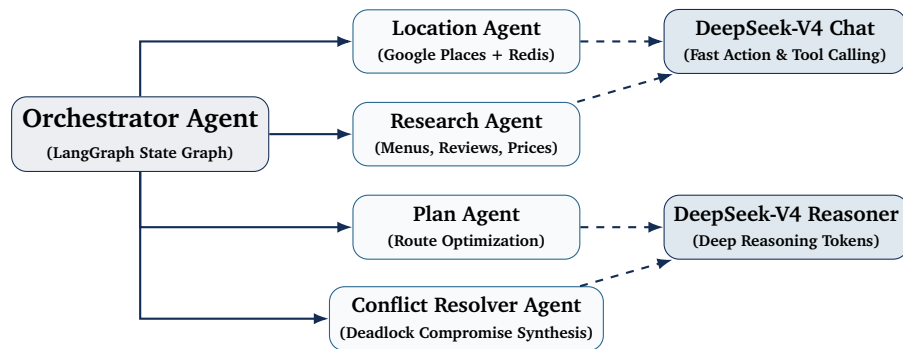


Figure 2: Orchestrator-Worker Multi-Agent Architecture with DeepSeek-V4

- **DeepSeek-V4 Advanced Reasoning Engine:** Employs reasoning tokens to solve complex combinatorial scheduling problems, respect spatial-temporal constraints, and analyze vote feedback to synthesize compromise solutions.

5. Key Technical Innovations

- **AI Conflict Resolver Agent:** When group votes are split (e.g., 3 votes for Japanese vs. 3 votes for BBQ), the agent analyzes members' comments and budget criteria to generate an optimal compromise (e.g., a venue serving both styles or re-sequencing the stops across dinner and late-night supper).
- **Real-Time Interactive AI Chat:** Members can brainstorm directly with the AI assistant via WebSocket/SSE streaming, instantly converting suggested locations into draft timeline cards.
- **Optimal 3-Table Debt Settlement Engine:** Eliminates messy calculations with a mathematically minimal 3-table schema (expenses, expense_splits, settlements). Net individual balance is computed directly:

$$\text{Net Balance}_u = \sum \text{Paid Amount}_u - \sum \text{Owed Share}_u \quad (1)$$

An automated greedy matching algorithm then calculates the minimum peer-to-peer transfers required to clear all balances.

- **High-Performance Integration Layer:** Redis caching layer guarantees sub-200ms location lookups and shields external APIs from quota limits.

6. Expected Impact & Value Proposition

- **Substantial Time Savings:** Slashes group planning time from 2–3 days of back-and-forth messaging down to **under 15 minutes**.
- **Democratic Alignment:** Combines public voting, transparent budgeting, and AI conflict resolution to prevent social friction and keep group outings enjoyable.
- **Technical & Scientific Value:** Serves as a state-of-the-art implementation of Multi-Agent LangGraph orchestration, Dual-Engine DeepSeek-V4 integration, and real-time collaborative web engineering.

TripMate AI transforms group planning from an exhausting chore into an enjoyable, collaborative experience.